Summer Internship Report

ON

Construction Practices In Tunnel Engineering

A Study at the Rishikesh – Karnprayag Rail Project (Package 2), Byasi

SUBMITTED BY

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Internship Duration

15 May 2026 to 14 July 2026

Internship Organization

Larsen & Toubro Limited

Rishikesh – Karnprayag Rail Project

Package – 2, Byasi

Uttarakhand



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Acknowledgement

I express my sincere gratitude to Larsen & Toubro Limited (L&T Construction) for providing me with the opportunity to undergo my two-month industrial internship at the Rishikesh–Karnaprayag Rail Project, Package-2, Byasi. This internship provided me with valuable practical exposure to tunnel construction and helped me understand the execution of large-scale infrastructure projects.

I would like to express my heartfelt thanks to Mr. Rajesh Chopra, Project Manager, for permitting me to undertake my internship at the project site. I sincerely thank Mr. Alok Kumar Patra and Mr. Paramjit Singh for their support, guidance, and coordination throughout my internship. I would like to express my sincere appreciation to Mr. Soumen Pal for his guidance and continuous support throughout the internship. I am especially thankful to Mr. Divyanshu Sharma for patiently explaining various construction activities and helping me understand practical engineering concepts.

I would also like to extend my sincere thanks to the entire staff of Adit-3 for their cooperation, encouragement, and willingness to share their practical knowledge and experience, which made my learning more meaningful throughout the internship.

Finally, I extend my heartfelt gratitude to everyone associated with the Rishikesh–Karnaprayag Rail Project, Package-2, who directly or indirectly contributed to making my internship a valuable and enriching learning experience. The guidance, support, and encouragement received during this period will remain a significant part of my professional learning and will always be remembered with gratitude.

CHAPTER 1: INTRODUCTION

Importance of Industrial Training

Civil Engineering involves various stages such as planning, design, construction, and maintenance of infrastructure including buildings, bridges, roads, tunnels, dams, and many other structures. In such a highly practical field, theoretical learning builds the foundation for becoming a good engineer, but practical exposure is equally important to understand how these concepts are implemented on site.

Industrial training provides students with an opportunity to visualize and understand what it really means to be a civil engineer. It helps in connecting classroom knowledge with real-life construction practices and gives a better understanding of the workflow followed during the execution of engineering projects. Through site exposure, students become familiar with construction methodologies, engineering drawings, structural components, construction materials, and modern equipment used in the industry.

Apart from technical knowledge, an internship also helps in understanding how different departments coordinate with each other to achieve a common objective. It teaches how real-life challenges are identified and solved, how decisions are taken at different stages of construction, and why proper planning, communication, quality control, and safety are essential for the successful completion of a project. Thus, industrial training plays an important role in developing the practical understanding and professional attitude required for a future civil engineer.

About the Company

The internship was carried out at Larsen & Toubro (L&T) Construction, one of India's leading engineering and construction organizations with a significant presence in infrastructure development. Established in 1938, L&T played an important role in the execution of several major projects in the fields of transportation infrastructure, buildings, heavy civil engineering, power, water, and industrial projects. The Heavy Civil Infrastructure division of L&T specializes in the execution of complex engineering projects such as metro systems, underground structures, tunnels, airports, highways, and railway infrastructure. The company adopts modern construction technologies along with stringent quality control measures and safety standards to ensure successful project execution. L&T emphasizes the principles of quality, safety, innovation, and sustainable development. The organization follows a systematic approach towards project planning, execution, monitoring, and risk management. Working in such a professional environment provides engineering students with valuable exposure to industry practices and modern construction techniques.

Internship Details

As a part of the B. Tech curriculum in Civil Engineering, a two-month industrial internship was undertaken at the Rishikesh–Karnaprayag Rail Project, Package-2, Byasi, from 15 May 2026 to 14 July 2026. To be more specific, I was posted at Adit-3, where I got the opportunity to gain practical exposure to various tunnel construction and quality control activities. The project involves challenging construction works, including long railway tunnels, underground structures, and associated civil engineering works being executed under difficult geological and topographical conditions. During the internship, I observed different stages of tunnel construction and learned how various construction activities are carried out in sequence. The major activities observed during the training included waterproofing layer installation, overt reinforcement fixing, secondary lining (overt) concrete using gantry, contact grouting, epoxy grouting, and gantry alignment. The internship provided valuable exposure to modern construction practices, quality control procedures, construction equipment, and the overall execution methodology followed in large-scale underground infrastructure projects.

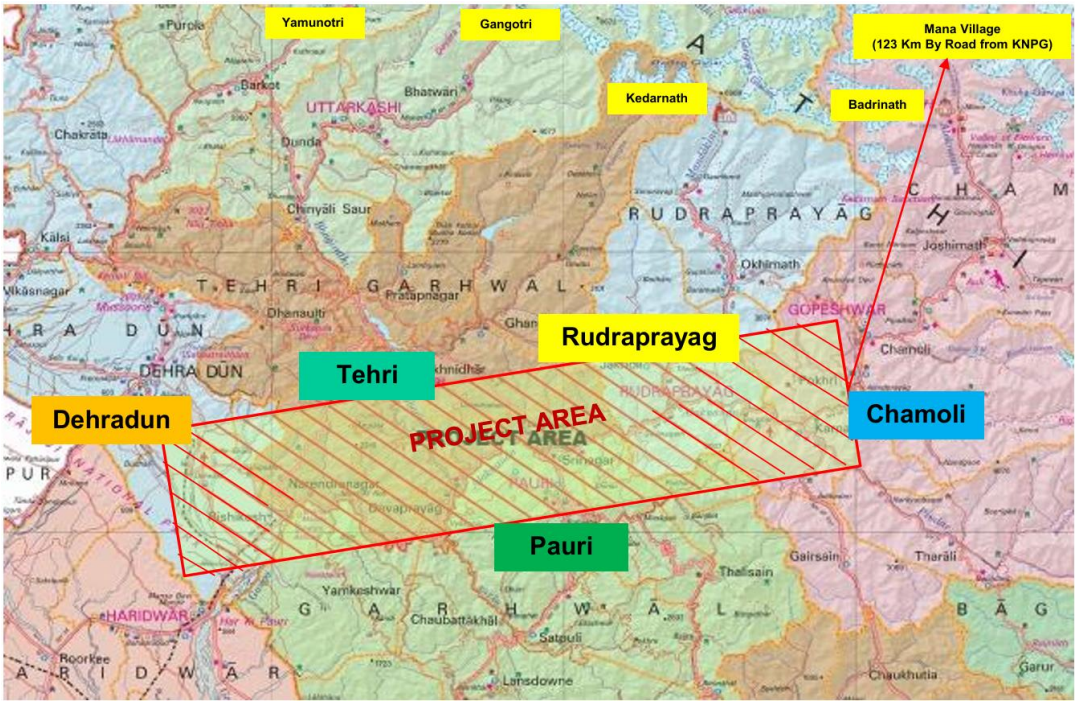
Objectives of the Internship

The internship was undertaken with the following objectives:

- To gain practical exposure to the execution of civil engineering works at a railway tunnel construction project.
- To observe different construction activities and understand their sequence, methodology, and practical implementation.
- To learn about the machinery, equipment, and construction techniques used during tunnel construction.
- To understand the importance of quality control procedures and safety practices at the construction site.
- To gain an understanding of the professional working environment and coordination among different departments involved in the project.
- To bridge the gap between the theoretical concepts studied during engineering and their practical application in the field.

Chapter 2: Project Overview

The Rishikesh–Karnaprayag Rail Link Project is one of the major railway infrastructure projects being executed in the state of Uttarakhand by Rail Vikas Nigam Limited (RVNL). The project involves the construction of a new 125 km Broad Gauge electrified railway line connecting Rishikesh with Karnaprayag. The alignment passes through the Himalayan terrain, where tunnels and bridges constitute a major portion of the railway line due to the difficult topography.



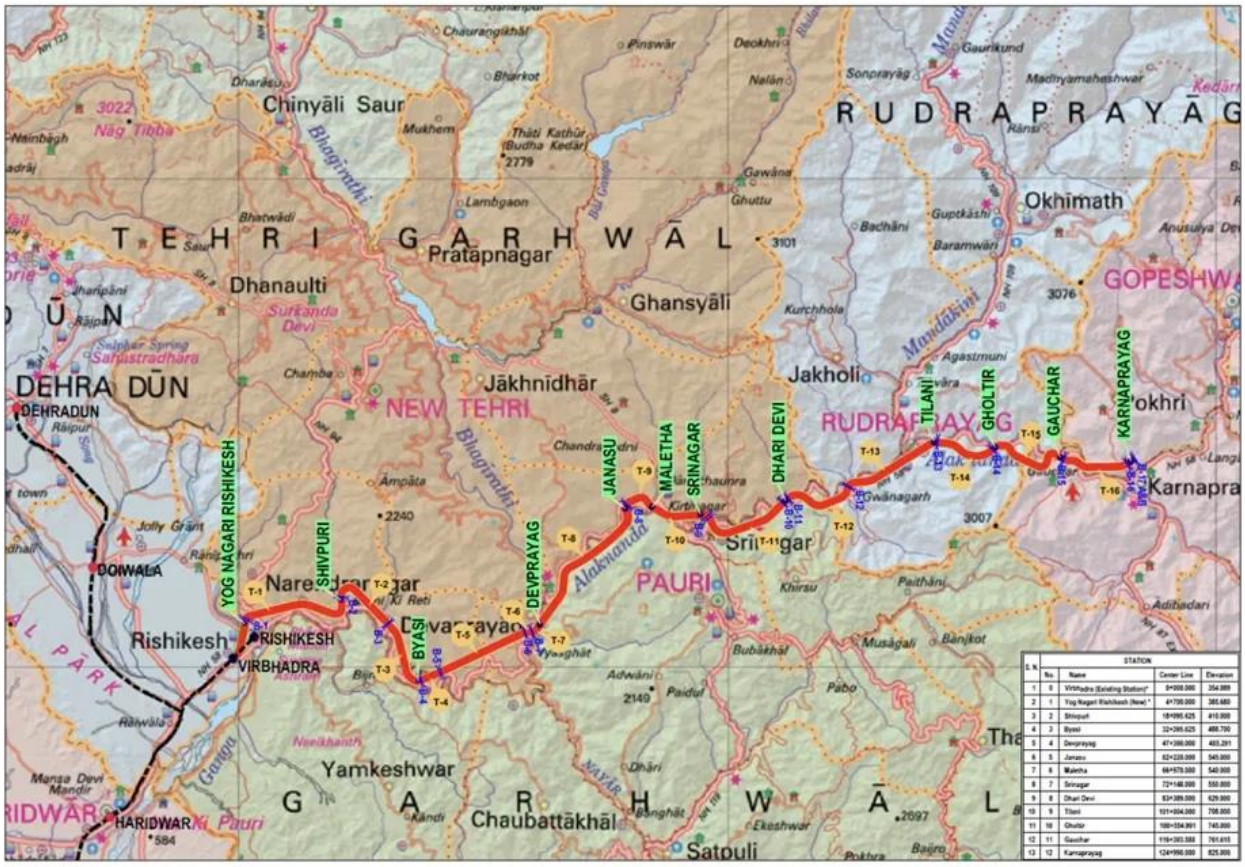
Particular	Details
Project Name	Rishikesh – Karnaprayag New Broad Gauge Rail Link Project
Package	Package – 2
Client	Rail Vikas Nigam Limited (RVNL)
Contractor	Larsen & Toubro Limited (Heavy Civil Infrastructure IC)
Consultant	Yuksel Proje & ICT (Joint Venture)
Project Location	Uttarakhand, India
Chainage	Km 18+444 to 33+097
Contract Value	₹ 1794.09 Crore
Construction Method	NATM (New Austrian Tunnelling Method)

Objectives of the Project

- To provide faster, safer, and more reliable rail connectivity between Rishikesh and Karnaprayag, improving access to major pilgrimage destinations such as Kedarnath and Badrinath.
- To promote regional economic growth by encouraging tourism, facilitating trade, and supporting the development of backward areas.
- To improve transportation infrastructure in the hilly terrain of Uttarakhand, reducing travel time and enhancing passenger convenience.
- To strengthen strategic connectivity in the Himalayan region by improving rail access towards the Indo–China border areas, supporting national security and future infrastructure development.
- To improve the quality of life in remote regions by enhancing accessibility, creating employment opportunities, and helping reduce migration.

Project Component	**Details**
Main Tunnel (T-2)	Length: 6090.5 m; Chainage: 18+685 to 24+775; Finished cross-sectional area: ≈45 m²
Escape Tunnel (T-2)	Parallel escape tunnel; Finished cross-sectional area: ≈18 m²
Main Tunnel (T-3)	Length: 6646 m; Chainage: 25+487.5 to 32+133.5; Finished cross-sectional area: ≈45 m²
Escape Tunnel (T-3)	Parallel escape tunnel; Finished cross-sectional area: ≈18 m²
Adit-2	Length: 605 m; Junction at 22+304
Adit-3	Length: 830 m; Junction at 27+772
Major Railway Bridges	2 Bridges (Bridge-2 at Chainage 18+448, Bridge-3 at 24+808), each having 4 × 30.5 m composite spans

ROUTE MAP OF THE PROJECT



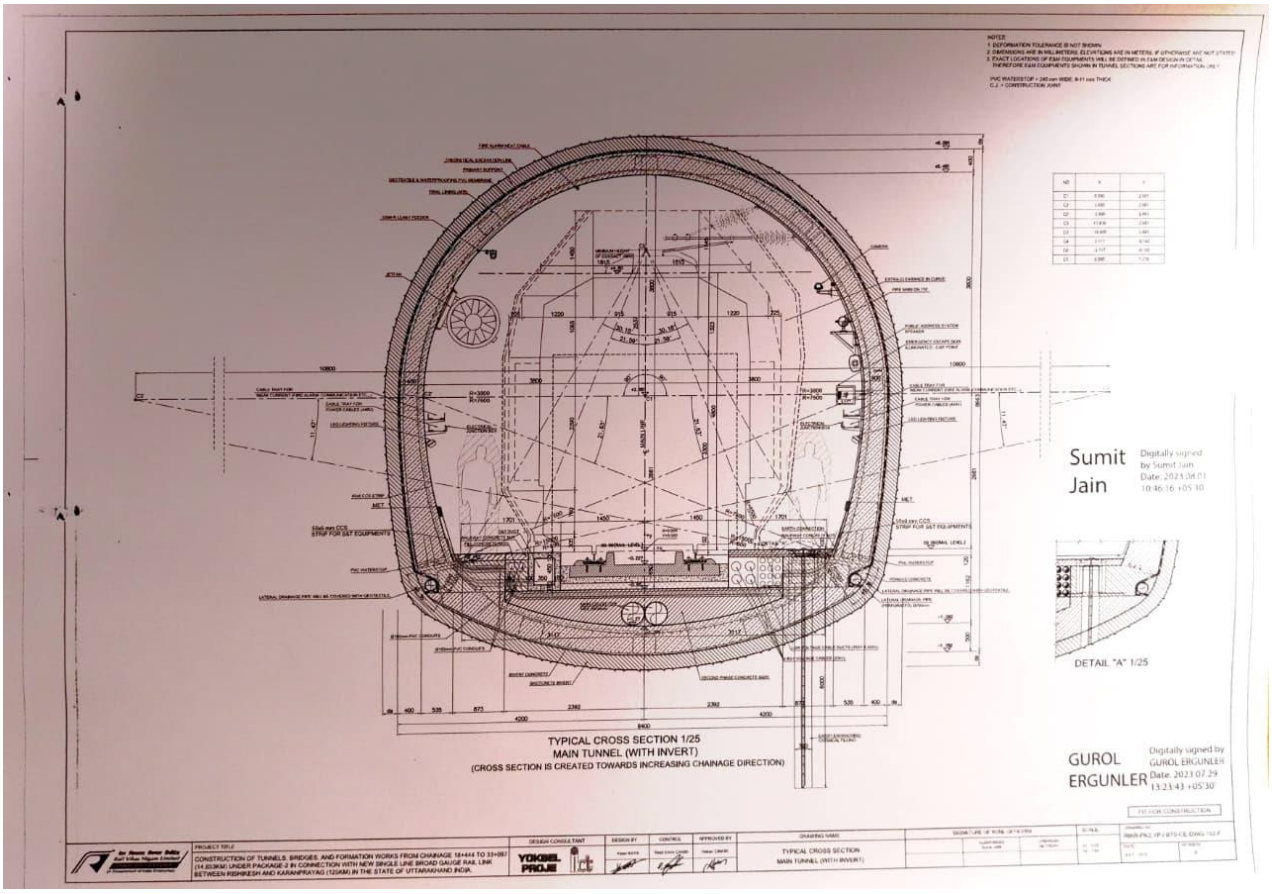
Chapter 3: Tunnel Components and Terminologies

Tunnel construction involves several structural components and auxiliary structures, each serving a specific function during construction as well as during the service life of the tunnel. Understanding these terminologies is essential for interpreting tunnel drawings and construction activities. This chapter briefly introduces the major tunnel structures used in the Rishikesh–Karnaprayag Rail Project and explains the important components of a typical tunnel cross-section.

Major Tunnel Structures

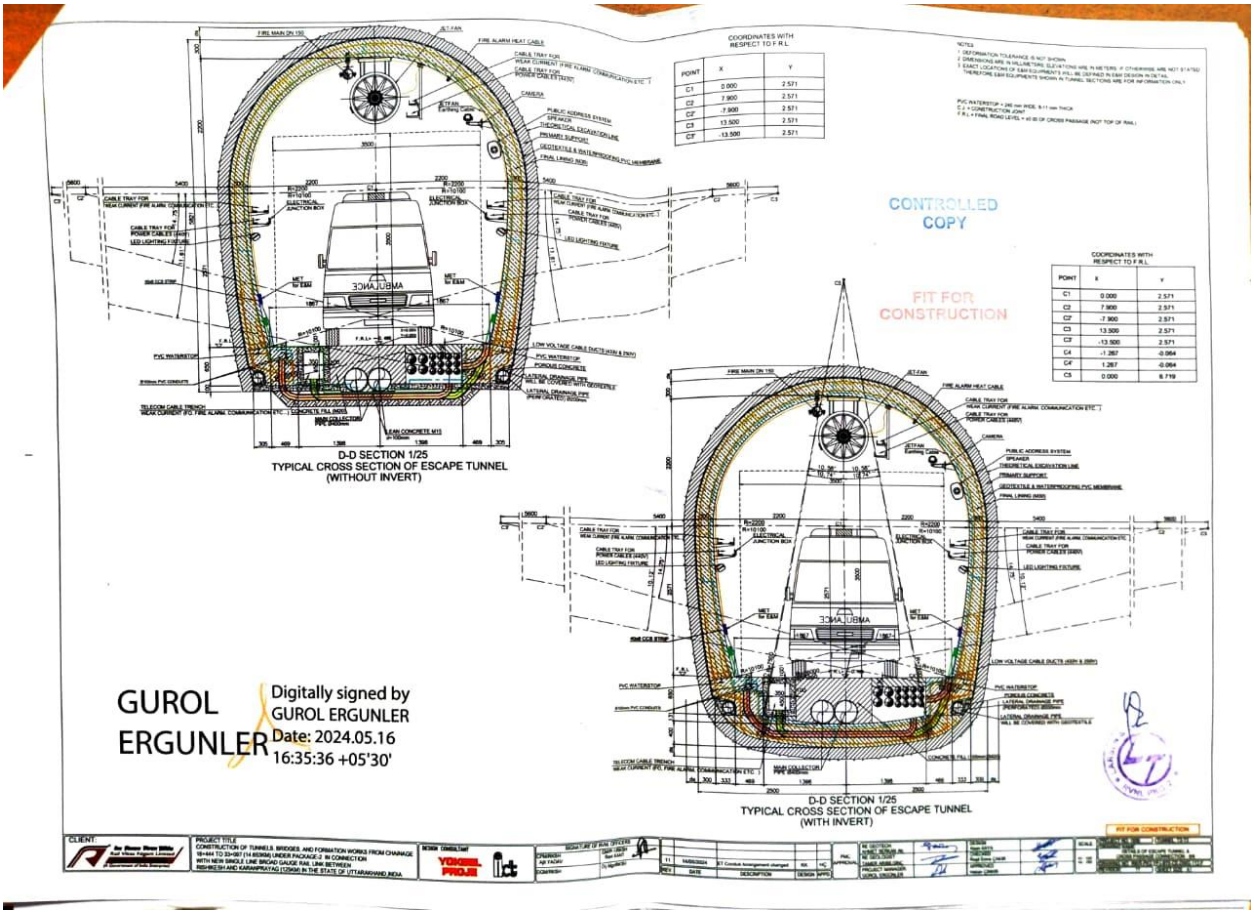
A. Main Tunnel

The Main Tunnel is the primary railway tunnel through which train movement takes place. It accommodates the railway track, overhead electrification, drainage system, cable ducts, walkway, and other operational facilities. In this project, the main tunnel forms the principal transportation corridor.



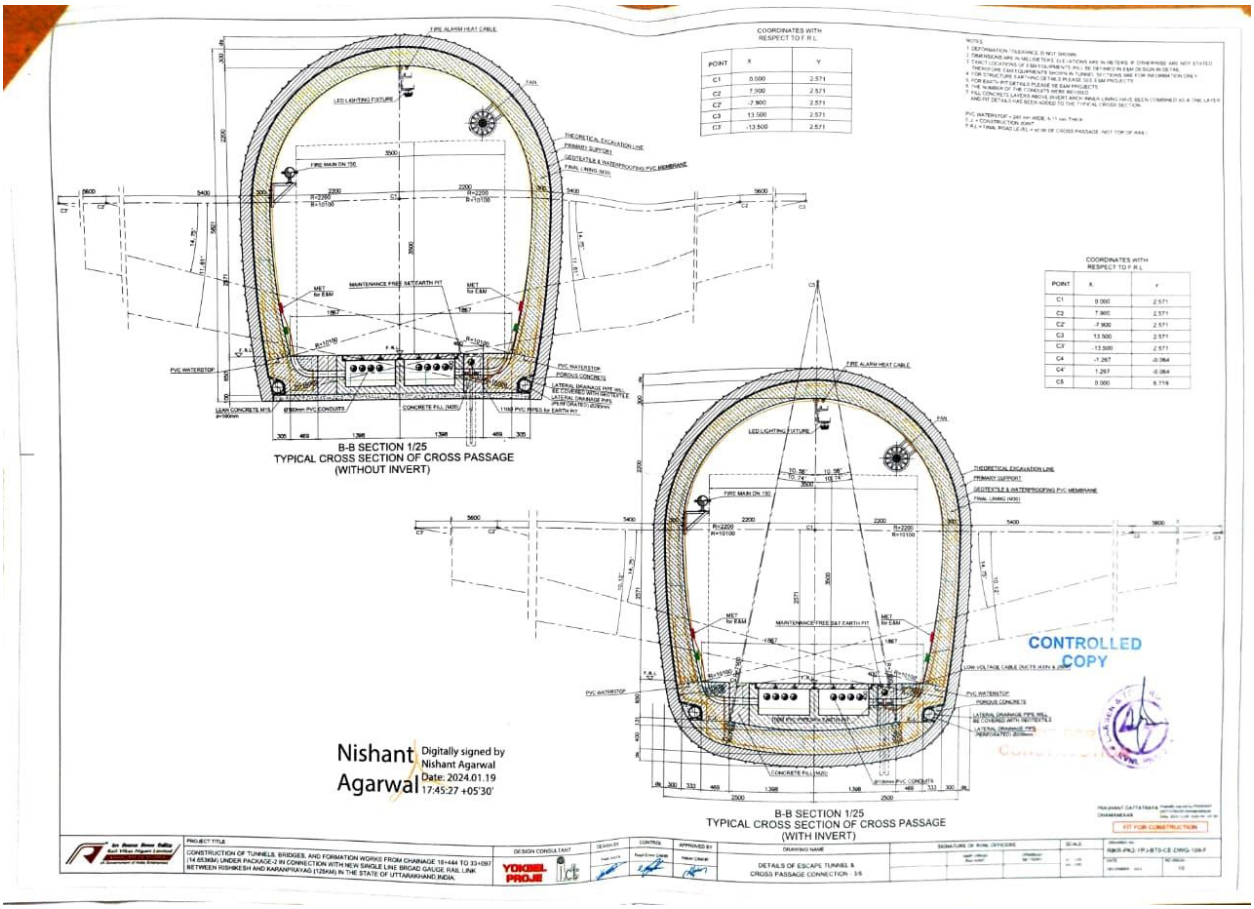
B. Escape Tunnel

The Escape Tunnel is constructed parallel to the main tunnel and serves as an emergency evacuation route. It provides safe passage for passengers and maintenance personnel during emergencies such as fire, smoke, or other incidents inside the main tunnel.



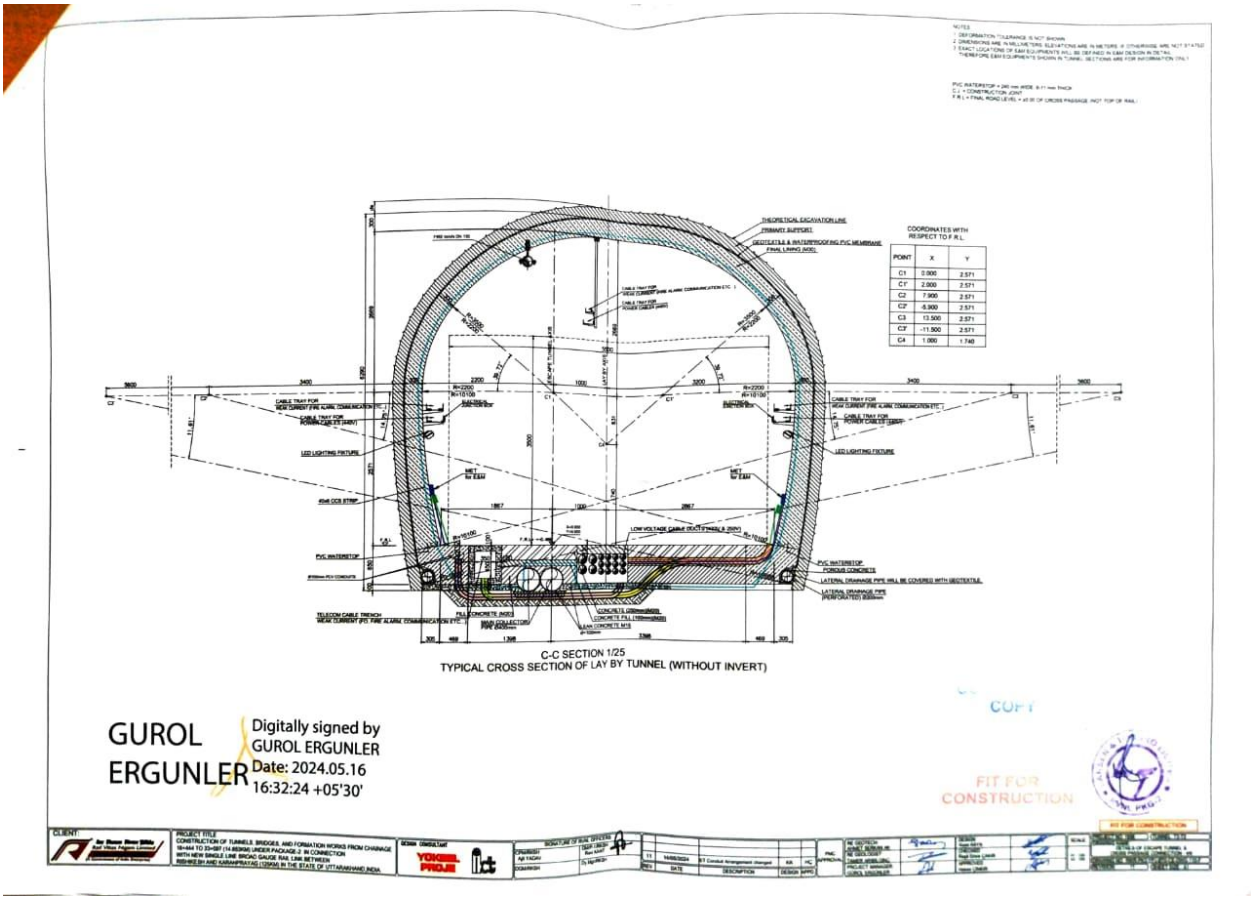
C. Cross Passage

A Cross Passage connects the main tunnel with the escape tunnel at regular intervals. It enables evacuation during emergencies and provides access for inspection and maintenance activities.



D. Lay-by

A Lay-by is an enlarged section provided within the tunnel to facilitate temporary parking of maintenance vehicles and storage of equipment. It also provides additional working space during maintenance operations.



E. Adit

An Adit is an access tunnel constructed from the ground surface to the main tunnel. During construction, it is used for transportation of manpower, machinery, materials, ventilation, and removal of excavated muck. After completion, it may also serve as an emergency or maintenance access depending on the project requirements.

General Features of Tunnel Cross Section

A typical tunnel cross section in the Rishikesh–Karnaprayag Rail Project consists of the primary support system, waterproofing system, and secondary support system. Together, these components provide stability, waterproofing, and long-term structural strength to the tunnel.

A. Tunnel Shape

The tunnels are constructed in a horseshoe-shaped cross section. This shape provides good structural stability underground pressure while offering sufficient space for railway tracks, drainage, cable ducts, walkways, and overhead electrification.

B. Primary Support System

The primary support is installed immediately after excavation to stabilize the surrounding rock and ensure safe working conditions. It consists of:

- Shotcrete – Sprayed concrete applied over the excavated surface for immediate support.
- Wire Mesh – Reinforcement provided within the shotcrete to control cracking.
- Lattice Girders – Steel ribs installed to provide additional structural support.
- SN Rock Bolts – Used for reinforcement of stable rock strata.
- SDA Rock Bolts – Self-drilling anchor bolts used in weak or fractured rock where conventional drilling is difficult.

C. Waterproofing System

The waterproofing system is installed between the primary and secondary support to prevent groundwater seepage. It consists of:

- 500 GSM Geotextile – Protects the waterproofing membrane from damage due to the rough shotcrete surface.
- 2 mm PVC Waterproofing Membrane – Acts as a watertight barrier, preventing water ingress into the tunnel.

D. Secondary Support System

The secondary support forms the permanent reinforced concrete lining of the tunnel and consists of:

- Reinforcement: 12 mm diameter bars arranged in a 150 mm × 150 mm grid.
- Concrete Cover: 75 mm cover provided for reinforcement.
- M30 Grade Reinforced Concrete: Cast using a tunnel lining gantry to form the final lining.

Components of a Typical Tunnel Cross Section

A. Overt

The Overt is the upper reinforced concrete portion of the secondary lining. It forms the roof and side lining of the tunnel and provides structural stability while resisting the loads transferred from the surrounding rock mass.

B. Invert

The Invert is the bottom portion of the tunnel lining. Its requirement depends on the ground conditions.

- With Invert: Reinforced concrete invert is provided where the rock mass is weak or unstable to complete the structural ring and improve stability.
- Without Invert: In competent rock, the invert may not be required, and the tunnel floor is constructed without reinforced concrete at the bottom.

A 75 mm concrete cover is maintained for the reinforcement in the reinforced concrete lining, including the invert where provided.

C. Kerb

The Kerb is a raised concrete element provided along both sides of the tunnel. It separates the central track zone from the walkway and provides support for drainage and cable duct arrangements.

D. Kicker

The Kicker is the base portion of the secondary lining from which the overt concrete is cast. It provides firm seating for the lining formwork (gantry) and ensures proper alignment during concrete.

E. Niches

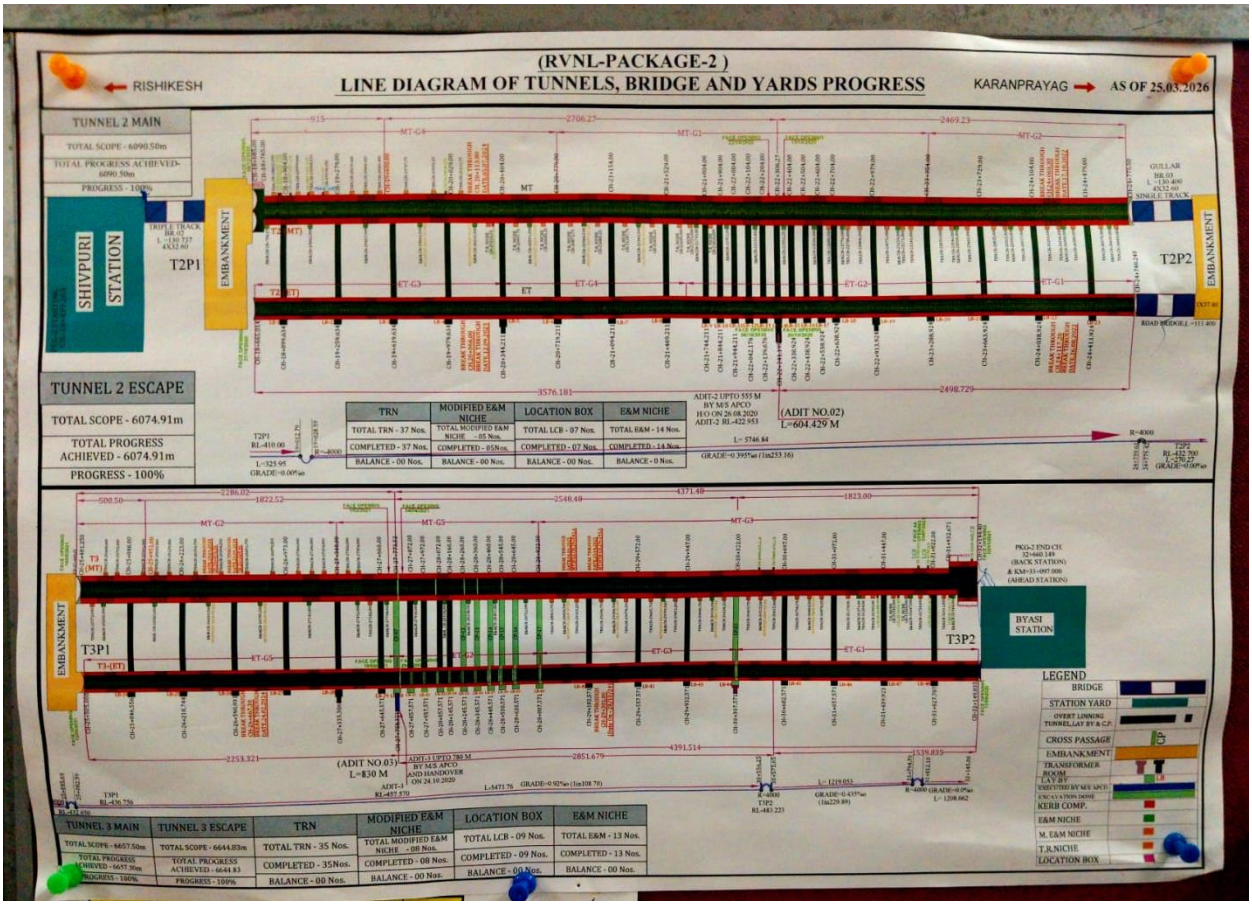
Niches are recesses provided at intervals along the tunnel wall. They provide safe standing space for maintenance personnel and can also accommodate emergency equipment such as fire extinguishers, communication systems, or electrical panels.

F. Drainage System

Proper drainage is essential to prevent water accumulation and hydrostatic pressure behind the tunnel lining. A 200 mm diameter perforated drainage pipe is installed behind the lining to collect seepage water from the surrounding rock. The collected water is discharged into a 400 mm diameter HDPE main collector pipe, which carries the water along the tunnel. Manholes are provided at suitable intervals to facilitate inspection, maintenance and cleaning of the drainage system.

G. Cable Ducts

Cable ducts are constructed along the sides of the tunnel to accommodate electrical, signalling, communication, and other utility cables. They protect the cables from mechanical damage while allowing easy inspection and maintenance.


(RVNL-PACKAGE-2)
LINE DIAGRAM OF TUNNELS, BRIDGE AND YARDS PROGRESS
RISHIKESH
KARANPRAYAG
AS OF 25.03.2026
TUNNEL 2 MAIN
TOTAL SCOPE - 6090.50m
TOTAL PROGRESS ACHIEVED- 6090.50m
PROGRESS - 100%
SHIVPURI STATION
T2P1
EMBANKMENT
T2P2
TUNNEL 2 ESCAPE
TOTAL SCOPE - 6074.91m
TOTAL PROGRESS ACHIEVED - 6074.91m
PROGRESS - 100%
TRN
MODIFIED E&M NICHE
LOCATION BOX
E&M NICHE
TOTAL TRN - 37 Nos.
TOTAL MODIFIED E&M NICHE – 05 Nos.
TOTAL LCB - 07 Nos.
TOTAL E&M - 14 Nos.
COMPLETED - 37 Nos.
COMPLETED - 05Nos.
COMPLETED - 07 Nos.
COMPLETED - 14 Nos.
BALANCE - 00 Nos.
BALANCE - 00 Nos.
BALANCE - 00 Nos.
BALANCE - 0 Nos.
(ADIT NO.02)
L=604.429 M
BYASI STATION
T3P1
T3P2
LEGEND
BRIDGE
STATION YARD
CROSS PASSAGE
EMBANKMENT
TRANSFORMER ROOM
KERB COMP.
E&M NICHE
M. E&M NICHE
T.R.NICHE
LOCATION BOX
(ADIT NO.03)
L=830 M
TUNNEL 3 MAIN
TUNNEL 3 ESCAPE
TRN
MODIFIED E&M NICHE
LOCATION BOX
E&M NICHE
TOTAL SCOPE - 6657.50m
TOTAL SCOPE - 6644.83m
TOTAL TRN - 35 Nos.
TOTAL MODIFIED E&M NICHE – 00 Nos.
TOTAL LCB - 09 Nos.
TOTAL E&M - 13 Nos.
TOTAL PROGRESS ACHIEVED - 6657.50m
TOTAL PROGRESS ACHIEVED - 6644.83
COMPLETED - 35Nos.
COMPLETED - 00 Nos.
COMPLETED - 09 Nos.
COMPLETED - 13 Nos.
PROGRESS - 100%
PROGRESS - 100%
BALANCE - 00 Nos.
BALANCE - 00 Nos.
BALANCE - 00 Nos.

Chapter 4: Activities Observed

Waterproofing Layer Installation

Activity

It involves installing a geotextile layer followed by a PVC membrane to provide a waterproof barrier between the primary support and the secondary lining.

Objective

- To prevent groundwater seepage into the tunnel.
- To protect the secondary lining and reinforcement from moisture and corrosion.
- To improve the durability and service life of the tunnel lining.

Materials, Machinery and Equipment Required

Materials

- 500 GSM Geotextile
- PVC Waterproofing Membrane (2 mm thick, 2100 mm wide)
- Cement mortar for surface padding
- PVC Rounders (Fixing Discs)

Machinery and Equipment

- Oxy-acetylene gas cutting set
- Drilling machine
- Hot air hand welding machine
- Twinny hot wedge welding machine (Leister/Weld-in)
- Working Platform (Jumbo)
- Measuring tape and marking tools

Procedure Adopted at Site

A. Surface Preparation

- The shotcrete surface was inspected for protruding rebars, steel wires and sharp projections.
- Such projections were removed using an oxy-acetylene cutting set.
- Uneven areas were levelled by applying cement mortar (padding) to obtain a smooth surface and prevent damage to the waterproofing membrane.

B. Installation of Geotextile

- A 500 GSM geotextile was installed over the prepared surface.
- It was fixed using PVC rounders anchored into the primary lining with a drilling machine.
- The geotextile acts as a cushioning layer, protecting the PVC membrane from puncture and helping in stress distribution.

C. Installation of PVC Membrane

- A 2 mm thick, 2100 mm wide PVC membrane was fixed over the geotextile.
- The membrane was locally welded to the rounders using a hot air hand welding machine (approximately 450°C to 600°C).
- Adjacent membrane sheets were laid with an overlap of about 100 mm.

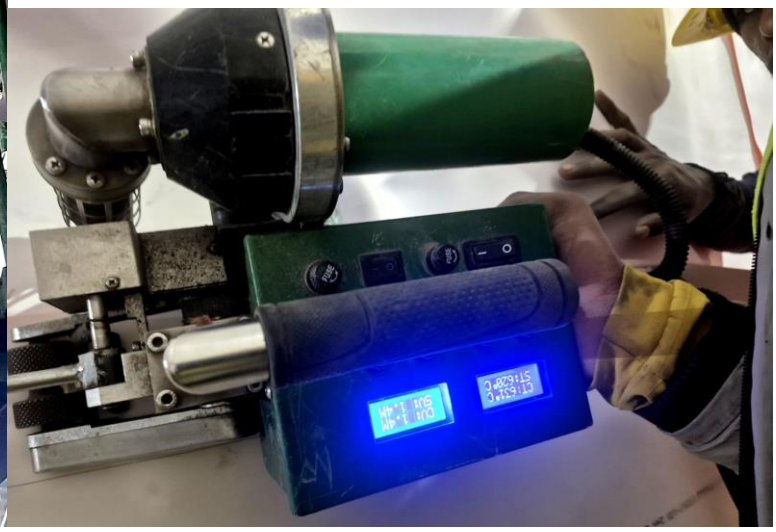
D. Double Seam Welding

- The overlapping membrane sheets were welded using a Twinny hot wedge welding machine (Leister/Weld-in).
- The machine produced two parallel welds, leaving an air channel between them for quality testing.
- Welding temperature, speed and overlap alignment were continuously monitored to ensure a uniform weld.

E. Quality Check

- After completion of welding, the welded joints were subjected to an Air Pressure Test to verify the integrity of the waterproofing system before reinforcement fixing and secondary lining concreting.





Gantry Alignment

Activity

Gantry alignment is a surveying activity carried out before overt concreting to position the tunnel lining gantry accurately at the required location. Proper alignment ensures that the secondary lining is cast according to the designed tunnel profile and centre line.

Objective

- To position the gantry at the correct chainage and alignment.
- To maintain the designed tunnel profile and dimensions.
- To ensure accurate positioning of the gantry before overt concreting.

Materials, Machinery and Equipment Required

- Total Station with Tunnel Profiler
- Survey Prism with Prism Pole
- Tripod
- Tunnel Lining Gantry
- Hydraulic Jacking System of Gantry
- Marking Paint

Procedure Adopted at Site

A. Instrument Setup

- The Total Station with Tunnel Profiler was set up inside the tunnel for alignment.
- Since the instrument could not be positioned directly over a known control point, the resection method was adopted.
- The instrument established its position by observing previously known survey control points.

B. Gantry Alignment

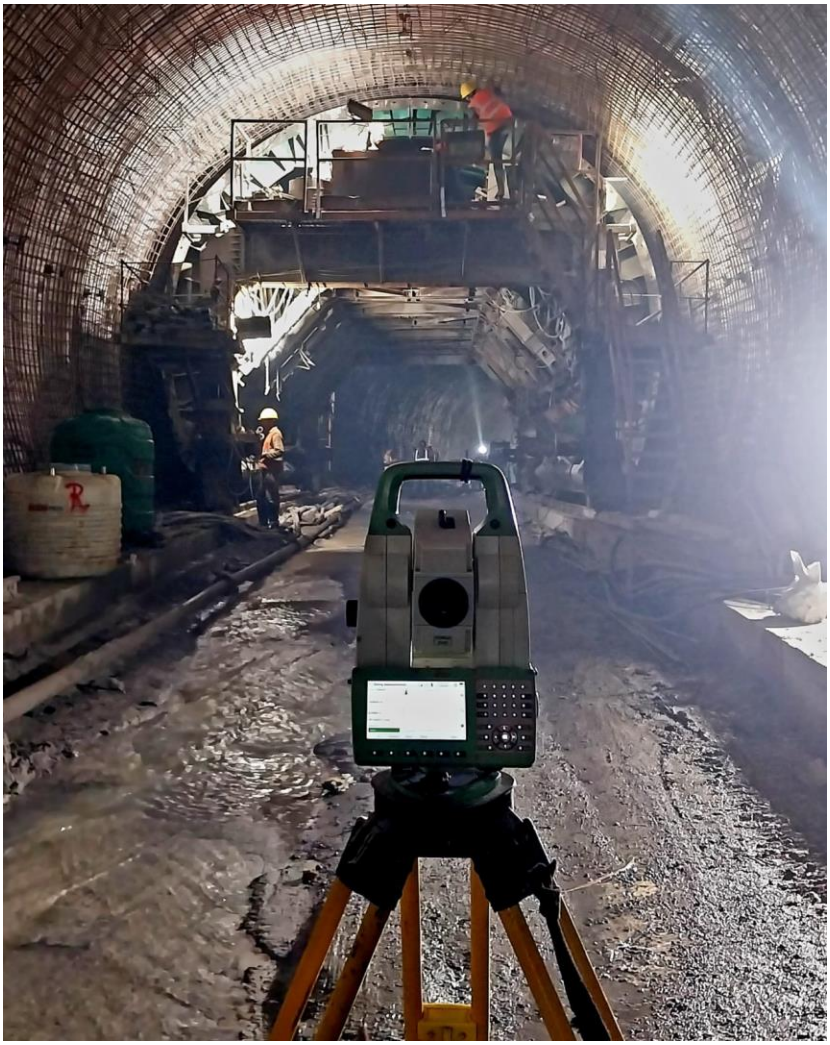
- The survey team referred to the alignment checklist provided, which contained the required coordinates for the gantry.
- A survey prism was placed successively at different reference points on the gantry, and observations were taken using the Total Station with Tunnel Profiler.
- During the observed activity, seven reference points were checked on one side of the gantry, comprising one point at the crown and three points on each side.
- The same alignment procedure was carried out on the opposite side of the gantry wherever concreting had not yet been completed.

C. Adjustment and Verification

- Based on the survey observations, minor adjustments were made using the gantry's hydraulic jacking system until the required position was achieved.
- The final alignment was verified before the start of overt concreting.
- This activity was repeated before every 12.1 m concreting cycle, corresponding to the length of one gantry panel.

Observations

- Gantry alignment was carried out by the Survey Team in coordination with the execution team.
- A Total Station with Tunnel Profiler and a survey prism were used for the alignment process.
- The resection method was adopted to establish the instrument position inside the tunnel.
- Alignment was carried out with a permissible tolerance of ±5 mm before approval for concreting.



Overt Concreting

Activity

Overt concreting is the process of casting the reinforced concrete secondary lining of the tunnel using a movable steel formwork called a lining gantry. At the observed site, a 12.100 m long gantry was used to cast each lining panel.

Objective

- To provide the tunnel with its final structural lining and designed profile.
- To protect the waterproofing system and improve the durability of the tunnel.
- To ensure a smooth internal surface suitable for railway operation.

Materials, Machinery and Equipment Required

Materials

- M30 Grade Ready Mix Concrete (RMC)
- 50 mm PVC contact grouting pipes
- Curing compound

Machinery and Equipment

- Tunnel Lining Gantry (12.100 m)
- Transit Mixers (TM)
- Concrete Pump and Delivery Pipeline
- Slump Test Apparatus
- Concrete Thermometer
- Cube Moulds
- Pendulum Rebound Hammer
- Wooden Bulkheads

Procedure Adopted for Overt Concreting

A. Gantry Preparation

- After approval of reinforcement, waterproofing and contact grouting pipes, the lining gantry was prepared for concrete.
- Wooden bulkheads were installed at both ends (if necessary) of the gantry to act as stop-end formwork and prevent leakage of fresh concrete.

B. Concrete Production and Transportation

- The concrete quantity required for one panel was estimated by the Quantity Surveyor.
- M30 Grade Concrete was produced at the batching plant located at Byasi.
- Concrete was transported to the tunnel using Transit Mixers then poured with the help of pump, pipelines, hoses.
- The complete concreting operation generally continued for 8–12 hours, depending on the concrete quantity.

C. Fresh Concrete Quality Checks

Before concrete placement, the Quality Control team carried out:

- Slump Test
- Concrete Temperature Test
- Cube Casting

Only after satisfactory test results were concrete permitted for placement.

D. Concrete Placement

- Concrete was pumped into the gantry through windows and nipples provided in the formwork.
- The pouring sequence was adjusted to achieve uniform filling of the side walls and crown.
- Unlike conventional concreting, needle vibrators were not used. Instead, the gantry was fitted with pneumatic surface vibrators, which compacted the concrete through the steel formwork.

E. Monitoring During Concreting

During concreting, the execution team continuously monitored:

- Concrete flows through the pouring nipples.
- Leakage from bulkheads and formwork joints.
- Appearance of cement slurry from designated locations, indicating complete filling.
- Continuous supply of concrete from the batching plant.

F. Post-Concreting Activities

- After completion of concreting, the concrete was allowed to gain initial strength inside the closed gantry.
- The wooden bulkheads were removed.

- An Early Strength Test was carried out before lowering the gantry.
- After achieving the required early strength, the gantry was lowered and shifted to the next panel.
- Finally, a curing compound was applied over the exposed concrete surface to minimize moisture loss.



Grouting

Grouting is the process of injecting specially prepared grout into voids, cracks, joints, or spaces within or behind tunnel linings under controlled pressure. It is carried out to eliminate voids, prevent water ingress, restore structural integrity, improve bonding between different construction elements, and enhance the long-term stability and service life of the tunnel. The type of grouting adopted depends on the nature of the defect or requirement at the site.

Contact Grouting

Activity

Contact grouting is carried out after completion of the secondary lining to fill any voids between the concrete lining and the waterproofing system by injecting cement grout through pre-installed grouting pipes.

Objective

- To fill voids behind the secondary lining.
- To improve contact between the lining and surrounding support.
- To minimize water seepage and enhance the durability of the tunnel lining.

Materials, Machinery and Equipment Required

Materials

- OPC 53 Grade Cement
- Clean Water

Machinery and Equipment

- Grout Mixer
- Grout Pump
- Pressure Gauge
- Flexible Grout Hose
- Packers
- Telehandler
- Hand Tools

Note: The grout mixer used at site had a capacity of 250 kg (5 cement bags) per batch.

Procedure Adopted at Site

A. Preparation

- During overt concreting, 50 mm diameter PVC pipes were embedded near the crown at approximately 4 m spacing to serve as contact grouting ports.
- Before grouting, the ports were inspected and cleaned to ensure they were free from obstruction.

B. Installation of Packers

- Packers were fixed at the grouting ports to provide a leak-proof connection with the grout hose.
- Since the ports were located near the crown, a telehandler was used for safe installation.
- The grout hose was then connected to the packers.

C. Preparation of Cement Grout

- Grout was prepared using OPC 53 Grade cement and clean water.
- A water-cement ratio of 0.40–0.50 was maintained.
- The grout was mixed in the grout mixer until a uniform consistency was achieved and then transferred to the grout pump.

D. Grout Injection

- Cement grout was injected through the embedded PVC pipes using the grout pump.
- Injection pressure was continuously monitored using a pressure gauge.
- Although the permissible pressure range was 0–5 bar, the operating pressure at site was generally maintained between 2.5 and 3.0 bar.
- Pressure was increased gradually to avoid overstressing the secondary lining.

E. Completion of Grouting

- Grouting was considered complete when a noticeable rise in pressure indicated that the voids had been filled.
- The quantity of grout consumed was recorded for each location and varied depending on the size of the voids.
- After completion, the grout hose and packers were removed before shifting to the next injection point.



Epoxy Grouting

Activity

Epoxy grouting is a repair technique used to seal water leakage through construction joints or minor cracks in the secondary lining. It involves injecting a two-component epoxy resin into the affected area under controlled pressure.

Objective

- To stop water seepage through construction joints and cracks.
- To restore the integrity of the concrete lining.
- To improve the durability and watertightness of the tunnel.

Materials, Machinery and Equipment Required

Materials

- Normet– Part A
- Normet – Part B
- Mixing Ratio: 1:1 by volume
- Renderoc Plug
- Thinner

Equipment

- Graco Ultra 390 Injection Machine
- Injection Hoses
- Mechanical Packers

- Hammer Drill with 15 mm Drill Bit
- Hand Tools
- Measuring Containers

Packers

The packers used during the observed activity were:

- Diameter: 12 mm
- Length: 100mm or 200mm

The packers consist of a rubber sleeve which expands on tightening, providing a leak-proof connection for resin injections.

Procedure Adopted at Site

A. Identification of Leakage

- Water leakage locations were identified, mainly near construction joints of the secondary lining.
- The affected locations were marked for drilling.

B. Drilling and Installation of Packers

- Holes were drilled on either side of the construction joint using a 15 mm drill bit.
- The holes were drilled at an inclined angle to intercept the leakage path.
- Mechanical packers were inserted and tightened, causing the rubber sleeve to expand and form a leak-proof seal.

C. Sealing of Construction Joint

- The construction joint and the area around the packers were sealed using Renderoc Plug.
- This prevented epoxy from escaping during injection and ensured proper penetration into the leakage path.

D. Preparation of Epoxy Resin

- Normet Part A and Part B were mixed in a 1:1 ratio by volume.
- The mixed resin had a working time of approximately 30–35 minutes.
- Thinner was used after completion of the work to clean the machine and hoses, preventing blockage.

E. Epoxy Injection

- The mixed resin was injected using the Graco Ultra 390 injection machine.
- Injection was carried out through the installed packers at a pressure of 0–1 bar.
- Grouting was continued until the resin no longer penetrated the joint and pressure increased, indicating that the voids had been filled.
- After completion, the packers were removed and the same procedure was repeated at the remaining leakage locations.


CONSTRUCTION



Chapter 5: Quality Control

Quality control is an essential part of tunnel construction, ensuring that materials and construction activities comply with the specified standards and project requirements. During my internship at ADIT-3, I observed several quality control tests carried out at different stages of construction. The major tests included the Air Pressure Test of PVC membrane welding, Slump Test, Cube Casting, and Early Strength Test using a Pendulum Rebound Hammer, all of which helped ensure the quality, durability, and safety of the tunnel lining.

Air Pressure Test of PVC Membrane Welding

Purpose

- To check the quality of welded joints of the PVC waterproofing membrane.
- To ensure there is no leakage before the final lining concrete is placed.

Equipment Used

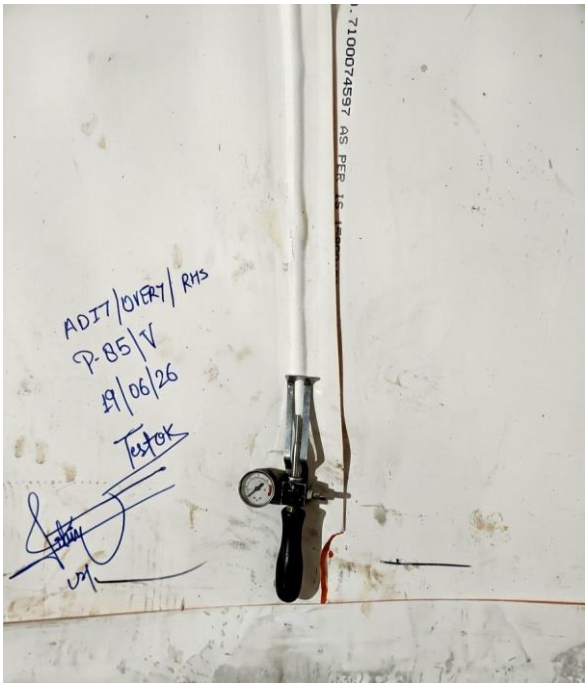
- Air Pressure Testing Kit with Pressure Gauge
- Hot wedge handheld welding machine

Procedure

- PVC membrane sheets were welded using a Twinny Double Seam Welding Machine, producing a 2 mm air channel between the two welds.
- Air was injected into the channel at a pressure of 2 bars.
- The pressure was maintained for 5 minutes.
- The pressure gauge was continuously monitored during the test.

Acceptance Criteria

- Pressure drop should not exceed 20% within 5 minutes.
- If the pressure remained within the permissible limit, the weld was accepted.
- Excessive pressure loss indicated leakage, requiring repair and retesting.



Slump Test

Purpose

- To determine the workability and consistency of fresh concrete before placement.
- To ensure only concrete meetings the specified workability is used for construction.

Equipment Used

- Slump Cone
- Tamping Rod
- Base Plate

Procedure

- The slump cone was placed on a rigid, level surface.
- Fresh concrete was filled into the cone in three equal layers.
- Each layer was compacted with 25 blows using a tamping rod.
- The cone was lifted vertically in a steady motion.
- The reduction in concrete height was measured as the slump value.

Observation

- Slump was checked for every transit mixer before entering the portal.
- Concrete grades generally used were M20 and M30.

- Required slump range: 160–200 mm.
- Typical observed slump during the internship: 190 mm.



Cube Casting and Compressive Strength Test

Purpose

- To determine the compressive strength of concrete.
- To verify that the concrete achieved the required design strength.

Frequency

- Cube casting was carried out according to the quantity of concrete poured, as per the project quality plan.

QTY of Concrete (CUM)	No. Of Cube Cast
5	6
15	9
30	12
50	15
100	18

+3 cubes are cast on every +50 m^3 concrete quantity.

Procedure

- Fresh concrete samples were collected during concrete.

- Concrete was placed into standard cube moulds in three layers.
- Each layer was compacted properly using a tamping rod.
- The top surface was finished and levelled.
- Each cube was labelled with project details such as:
    - ADIT-3 / Adit Overt
    - Concrete Grade (e.g., M30A20)
    - Pour Number (e.g., P-48)
    - Date of Casting
- Cubes were cured at the construction site.
- After curing, they were transported to the laboratory at Byasi for compressive strength testing.

Testing

- Compressive strength tests were conducted at:
    - 7 Days
    - 28 Days



Early Strength Test using Pendulum Rebound Hammer (NDT)

Purpose

- To determine the early strength of concrete for deciding whether the gantry could be safely lowered after concreting.

Equipment Used

- Pendulum Rebound Hammer

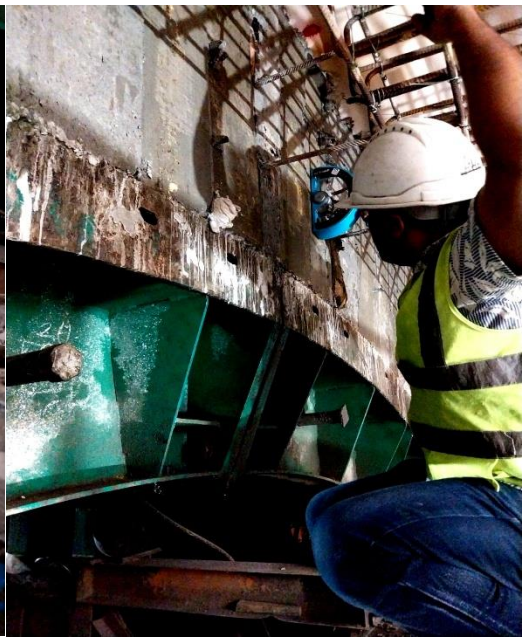
Procedure

- Testing began approximately 8 hours after completion of concreting.
- If the required strength was not achieved, the test was repeated every 1 hour.
- The concrete surface was first smoothened.
- Testing was carried out at the crown of the tunnel lining, as it is the last portion to gain strength.
- The rebound hammer was held in the vertical direction.
- Nine readings were taken from 3–4 different locations.
- The rebound (R) values were converted into equivalent cube compressive strength using the standard conversion formula.
- The average compressive strength was calculated.

Acceptance Criteria

- For Adit Overt, the average early strength should be 10–11 MPa.
- If the average strength reached the required value, the gantry was permitted to be lowered.
- If not, the test was repeated after one hour until the required strength was achieved.





Chapter 6: Environment Health & Safety

Safety is the foremost priority in any tunnel construction project due to the presence of heavy machinery, confined spaces, underground operations, and continuous construction activities. My first experience of the project reflected this strong safety culture.

As soon as I joined the project at Byasi, I was first introduced to the Safety In charge, who briefed me about the basic safety rules, emergency procedures, and the precautions to be followed while working inside the tunnel. Only after completing this safety induction was, I issued my Personal Protective Equipment (PPE), which included a reflective safety jacket, safety helmet, safety shoes, and gumboots. After ensuring that I was properly equipped and aware of the safety requirements, I was permitted to visit ADIT-3. This clearly demonstrated that at L&T, safety comes before any construction activity.

Safety Philosophy and Zero Tolerance Policy

L&T Construction follows a strong safety philosophy with the objective of achieving Zero Harm. The company believes that all accidents are preventable and that no work is so important that it should compromise the safety of workers or the environment.

The project also follows a Zero Tolerance Policy, under which serious violations of safety rules are not accepted. Every engineer, supervisor, worker, and visitor is expected to always follow the prescribed safety procedures.

Personal Protective Equipment (PPE)

The use of PPE was compulsory for every person entering the construction site. The standard PPE observed during my internship included:

- Safety Helmet
- Reflective Safety Jacket
- Safety Shoes
- Gumboots (where required)
- Safety Gloves
- Safety Goggles
- Ear Protection for high-noise activities
- Safety Harness while working at heights

The proper use of PPE minimizes the risk of injuries and ensures a safer working environment for everyone on the project.

Life Saving Rules (LSRs)

L&T implements **11 Life Saving Rules (LSRs)**, which are mandatory for everyone working on the project. These rules help prevent major accidents and promote safe working practices.

The eleven Life Saving Rules are:

1. Fit for Duty
2. Risk Management
3. Work Authorization
4. Safe Driving
5. Energy Isolation
6. Confined Space
7. Hot Work
8. Line of Fire
9. Safe Lifting
10. Working at Height
11. No Bypassing of Safety Controls

EHS Standards and Safe Work Practices

The project follows several Construction EHS Standards and IMS Procedures to ensure safe execution of work. These include standards related to working at height, excavation, electrical safety, confined space entry, plant and equipment safety, tunnel safety, housekeeping, and waste management.

Before carrying out any high-risk activity, a Risk Assessment is conducted, and a Work Permit is issued wherever required. This ensures that hazards are identified in advance and suitable preventive measures are taken.

Chapter 7: Conclusion

By observing various construction activities during my internship, I learned many practical aspects of civil engineering and was able to connect the theoretical concepts studied during my engineering courses with real-life construction practices. This experience helped me understand that engineering is not just about memorizing concepts but also about visualizing their practical application and understanding the importance of every procedure that we usually studied only in laboratory reports and textbooks. During the internship, I observed activities such as gantry alignment, waterproofing layer installation, overt concreting, contact grouting, and epoxy grouting. Each of these activities strengthened my understanding of engineering concepts and demonstrated how theoretical knowledge is implemented on site to ensure quality, safety, and durability in tunnel construction. These technical learnings were valuable and were expected from an industrial internship. However, apart from these, I have three important learnings that I will carry with me throughout my professional career simply by observing the working environment at the construction site. Their importance extends beyond engineering and is applicable to every profession.

1. Importance of Safety The first and most important learning was the importance of safety—not only at a construction site but in every workplace. I realized that no work is more important than human life, and this principle was reflected in every safety procedure followed at the site. From the mandatory use of PPE to safe working practices, it became clear that productivity should never be achieved at the cost of safety.

2. Importance of Process and Methodology Another important learning was understanding the importance of following the correct process and methodology. We generally know what materials are required and where they are used, but I learned that the actual strength, durability, and integrity of a structure depend equally on how the work is executed. Even small details in construction procedures can significantly affect the quality of the final structure. Earlier, while writing laboratory reports in college, I never fully understood the importance of lengthy procedures, but after observing real construction activities, I now appreciate why every step must be followed carefully.

3. Importance of Communication The third learning was the importance of communication. Communication is the key to the successful execution of a large construction project. Such projects are never the effort of a single person but the result of coordinated work among multiple teams and departments. I realized that every individual's work affects the work of others, and therefore timely communication and coordination are essential. Informing others about ongoing activities, potential issues, and possible solutions helps ensure smooth project execution. This

learning is not limited to construction projects but is applicable in every professional environment.

Finally, I would like to conclude by saying that this internship allowed me to experience the complete process of observing, learning, and understanding real construction practices. It also made me realize how significantly a civil engineer contributes to society by developing infrastructure that provides better facilities and improves people's quality of life. Moreover, civil engineers create structures that become a source of pride for the nation. This internship has further strengthened my interest in Civil Engineering, and I look forward to contributing to the development of safe, durable, and sustainable infrastructure in the future.